Pediatric Diabetes 2008: 9: 53–59
doi: 10.1111/j.1399-5448.2007.00258.x
All rights reserved

© 2007 The Authors
Journal compilation © 2007 Blackwell Munksgaard

Pediatric Diabetes

Original Article

Prevalence of metabolic markers of insulin resistance in offspring of gestational diabetes pregnancies

Keely EJ, Malcolm JC, Hadjiyannakis S, Gaboury I, Lough G, Lawson ML. Prevalence of metabolic markers of insulin resistance in offspring of gestational diabetes pregnancies.
Pediatric Diabetes 2008: 9: 53–59.

Abstract: *In utero* hyperglycemia has been associated with insulin resistance (IR) in children; however, there are limited data in low-risk populations. The purpose of this study was to describe the prevalence of metabolic markers of IR in a primarily Caucasian cohort of gestational diabetes mellitus (GDM) offspring aged 7–11 yr (mean 9.1) and to correlate offspring with maternal indexes. Sixty-eight children were recruited through a follow-up study of women who participated in a randomized controlled trial of minimal intervention vs. tight glycemic control for GDM. All participants had a fasting plasma glucose (FPG), insulin, total cholesterol, high-density lipoprotein cholesterol (HDL-chol), triglyceride (TG) level, and a 2-h oral glucose tolerance test. We calculated homeostasis model assessment (HOMA) and recorded body mass index and waist circumference (WC). Criteria for metabolic syndrome for children included: FPG $>$ 6.0 mmol/L, HDL-chol $<$ 1.03 mmol/L, TG $>$ 1.24 mmol/L, WC $>$ 90% for age and gender, and 2-h glucose $>$ 7.8 mmol/L. Among these children, 45 (66%), 17 (25%), 5 (7%), and 1 (1.5%) had zero, one, two, or three metabolic markers of IR, respectively. Hypertriglyceridemia (21%) was most prevalent, with no child having an elevated FPG. WC ($p = 0.018$) and TG ($p = 0.005$) were strong predictors of IR in the offspring after adjustment for age, gender, birthweight, family history, and maternal IR. Maternal and offspring HDL-chol, TG, WC, and HOMA but not fasting or 2-h glucose levels were significantly correlated. We conclude that metabolic markers of IR in children exposed to GDM may be present in the absence of abnormal fasting or 2-h glucose values. Screening strategies that focus on glucose levels may need to be reconsidered to institute early intervention with lifestyle changes for children at risk.

Erin J Keely[a], Janine C Malcolm[a], Stasia Hadjiyannakis[b], Isabelle Gaboury[c], Gigi Lough[d] and Margaret L Lawson[b]

[a]Division of Endocrinology and Metabolism, University of Ottawa, Ottawa Hospital, Riverside Campus, Ottawa, Ontario, Canada; [b]Division of Pediatrics, University of Ottawa, Children's Hospital of Eastern Ontario, Ottawa, Ontario, Canada; [c]Chalmers Research Group, Children's Hospital of Eastern Ontario, Ottawa, Ontario, Canada; [d]Children's Hospital of Eastern Ontario, Ottawa, Ontario, Canada

Key words: gestational diabetes – IR – offspring

Corresponding author:
Dr Erin J Keely
Division of Endocrinology and Metabolism
University of Ottawa
Ottawa Hospital, Riverside Campus
1967 Riverside Drive
P.O. Box 215, Ottawa, ON
Canada K1H7W9.
Tel: 613-738-8400/x81946;
fax: 613-738-8396;
e-mail: ekeely@ottawahospital.on.ca

Submitted 10 August 2006. Accepted for publication 15 May 2007

The increasing prevalence of type 2 diabetes mellitus (T2DM) and obesity in children and adolescents is a growing concern (1, 2). The metabolic syndrome, which is closely associated with insulin resistance (IR), is defined by clinical and laboratory features of abdominal obesity, hypertension, atherogenic dyslipidemia, and glucose intolerance (3–6). In adults, these components are predictive of cardiovascular disease and T2DM. Features of the metabolic syndrome are being increasingly identified in children. IR in childhood likely predicts an increased lifetime risk of atherosclerosis and diabetes as the features of metabolic syndrome continue from childhood to adulthood (7–9).

Maternal diabetes reflects a metabolically altered fetal environment where maternal hyperglycemia leads to excess fetal insulin production. *In utero* exposure to hyperglycemia particularly in high-risk groups such as Pima Indians, First Nations, and African

American children has been associated with IR in offspring (10–15). There are limited data on this association in low-risk populations.

The development of cost-effective screening strategies for early detection of children at risk is important for targeting susceptible individuals for weight control and risk factor surveillance. The Canadian Diabetes Association 2003 Clinical Practice Guidelines recommend screening for T2DM during childhood only in children who are >10 yr of age, obese, and meet two other criteria including high-risk ethnic group, family history of T2DM, acanthosis nigricans, polycystic ovary disease, hypertension, or dyslipidemia (16). Because offspring from pregnancies complicated by gestational diabetes mellitus (GDM) show evidence of IR at a young age (12), they may be a group to target with earlier, more aggressive screening.

We had previously conducted a randomized controlled trial of women with GDM from 1991 to 1995 (17). The intent of this original study was to compare tight glycemic control for GDM vs. minimal intervention on perinatal outcomes. This cohort of women provided the opportunity to study the long-term outcomes of GDM on offspring in a low-risk population. We have previously reported on the increased incidence of glucose intolerance in this offspring cohort (18).

The purpose of this study was to describe the prevalence of metabolic markers of IR in this low-risk Caucasian cohort of GDM offspring at 7–11 yr of age. A secondary objective was to determine if maternal factors correlate with offspring risk of markers of IR.

Methods

Subjects and protocol

A 7- to 11-yr follow-up study was conducted of the 299 women (91% Caucasian) who participated in the original randomized controlled trial that compared tight glycemic control and minimal intervention in the management of GDM (17). Attempts were made to contact all families using letters and telephone calls. The study was set at the Ottawa Hospital (General and Civic campuses) and the Children's Hospital of Eastern Ontario, Ottawa, Canada. Ethics approval was obtained from the research ethics boards of the Ottawa Hospital and the Children's Hospital of Eastern Ontario. As this was a follow-up study, sample size was dependant on the participation rate.

Women were originally screened for GDM by giving a 75-g glucose screening test between 24 and 28 wk gestation, with a 1-h postscreen cutoff of 8.0 mmol/L (17). Women with a positive screening test result underwent an oral glucose tolerance test with a 75-g glucose load. Diagnosis of GDM was made if 2-h glucose level was >9.6 mmol/L as per the criteria of Hatem et al. (19). The criteria used for screening and diagnosis of GDM in this study are different from current practice guidelines (16) but are reflective of the clinical practice from 1990 to 1995 at the Ottawa Hospital. Women were randomized to one of two treatment groups – *tight control group* [followed biweekly in a tertiary care center with glycemic targets of fasting blood glucose (FBG) of <4.4 mmol/L (<80 mg/dL) and 1-h postprandial blood glucose (BG) of <7.8 mmol/L (140 mg/dL)] or *minimal intervention group* [followed by their primary obstetric care provider, self-monitored glucose result performed biweekly, phoned to a research assistant with transfer to the treatment arm if they had persistent FBG of >7.8 mmol/L (140 mg/dL) or 1-h postmeal BG > 11.1 mmol/L (>200 mg/dL)]. There were no significant differences in perinatal outcomes of macrosomia, birth trauma, or hypoglycemia between the two original treatment groups (17).

Each participant (mothers and offspring) stood in bare feet, with heels and toes together with their back against the stadiometer, which was attached to the wall for height measurement. Heads were positioned so that the tragus of the ear and the corner of the eye made a horizontal line. Weight was measured on an electronic scale. Body mass index (BMI) (kg/m^2) was calculated for each participant. Three measurements of waist circumference (WC), using a tape measure placed at the midpoint between the top of the iliac crest and the bottom of the rib cage, were averaged.

All participants had a plasma glucose, insulin, total cholesterol, high-density lipoprotein cholesterol (HDL-chol), and triglyceride (TG) level performed after an 8-h fast. Two hours after a 1.75-g/kg glucose load, to a maximum of 75 g, was ingested, plasma insulin and glucose levels were drawn. Blood pressure was not measured. Participants already diagnosed with diabetes were included in the study but did not undergo the 2-h glucose tolerance test. Ethnicity and family history of diabetes were obtained through a self-report questionnaire given to mothers of the offspring. Tanner staging was not assessed.

Definitions

A uniform definition for the metabolic syndrome in the pediatric population does not exist. Given that many of the features of the metabolic syndrome track from childhood into adulthood, the definition should include similar components as those defined for the adult population. The criteria used in this article were therefore adapted from the adult National Cholesterol Education Program ATP III (20). Cutoffs were as follows: FPG > 6.0 mmol/L (impaired fasting glucose), HDL-chol < 1.03 mmol/L (<5% for age), TG > 1.24 mmol/L (>90% for age), WC > 90% for age and gender (21), and 2-h glucose value > 7.8 mmol/L [impaired glucose tolerance (IGT)]. Although IGT is

13995448, 2008, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1399-5448.2007.00258.x. By Cochrane Peru, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

not included in the adult definition of metabolic syndrome, we believed that it was important to include it as a criterion with respect to pediatric subjects. Impaired fasting glucose is very rare in children, even those at high risk, compared with glucose intolerance (22, 23). In fact, impaired fasting glucose in children may be a late manifestation of abnormal glucose homeostasis. WC was chosen over BMI as visceral fat is more closely associated with IR and cardiovascular risk (24, 25). Children were defined as at risk of overweight if their BMI (kg/m^2) was greater than the 85th percentile but less than the 95th percentile for age and sex and as overweight if BMI greater than the 95th percentile as defined by the 2000 Center for Disease Control and Prevention growth charts for the United States (26). Criteria used to define metabolic markers associated with IR in children are not well defined (27). IR was calculated using the homeostasis model assessment (HOMA) estimate of insulin sensitivity method [fasting insulin (mU/mL) multiplied by fasting glucose (mmol/L) divided by 22.5], which has been shown to be a valid marker of insulin sensitivity in children and adolescents (28). Higher values indicate greater IR. A cut point for children has not been clearly defined, so we elected to use the adult cut point of 2.5.

Analytical methods

Glucose was measured on the Vitros analyzer (Johnson and Johnson Clinical Diagnostics, Rochester, NY, USA) using manufacturer's reagents (glucose oxidase method). Insulin was measured on the Beckman Coulter Access 2 (Fullerton, California). Total cholesterol and triglycerides were measured on the Vitros analyzer. HDL-cholesterol was analyzed on the Beckman Coulter LX20 PRO (Brea, California, USA). Low-density lipoproteins were calculated using the Friedewald equation.

Statistical analysis

Descriptive statistics were generated. The skewness and kurtosis for the variables were assessed using both statistical and graphical methods. Medians rather than means were reported as a result of deviations from normality. Predictors of increasing IR as measured by HOMA were determined using a multivariate linear regression model examining a number of independent variables and adjusted for child's age, gender, family history of diabetes, and presence of metabolic syndrome in the mother. Crude and adjusted coefficients of regression were obtained along with their 95% confidence interval. The Wilcoxon Mann–Whitney test was used to determine the relationship between metabolic markers of IR in the offspring and offspring birthweight and between presence of metabolic markers in the offspring and maternal glucose control during pregnancy. The relationship between the presence of markers for metabolic syndrome among large-for-gestational-age (LGA) babies, average-for-gestational-age (AGA) babies and pre- and postpreprandial glucose control was determined by chi-squared or Fisher's exact tests where appropriate. Associations between maternal characteristics and metabolic syndrome markers were determined by Pearson correlation coefficients.

Results

We were successful in recruiting 30% of the original cohort (89 of the 299 patients). Not all children who participated agreed to have blood work and/or completed the 2-h glucose tolerance test. Data reported are only for those with a complete data set (Fig. 1).

Characteristics of the participating mothers and offspring are shown in Table 1. There was no significant difference in mean age at delivery ($p = 0.26$), mean gestational length ($p = 0.26$), prepregnancy weight ($p = 0.74$), or delivery weight ($p = 0.83$) between the group that chose to participate and the group that did not (data not shown).

There were 45 (66%), 17 (25%), 5 (7%), and 1 (1.5%) child(ren) with zero, one, two, or three metabolic markers of IR. Table 2 shows the prevalence of each marker. Of the 68 children, 49 (72%) had normal BMI, 8 (12%) were overweight (BMI between 85th and 95th percentile), and 11 (16%) were obese. Eight (16%) of the 49 offspring of normal BMI had one marker of metabolic syndrome. Six (75%) of the eight overweight offspring had at least one marker of metabolic syndrome (three with one marker and three with two markers). Of the 11 obese offspring, 9 (82%) had markers of metabolic syndrome (six with one marker, two with two markers, and one with three markers).

Using a multivariate linear regression model, we found that WC > 90th percentile (beta: 0.932, 95% CI: 0.236–1.629, $p = 0.010$) and TG > 1.24 mmol/L (beta: 0.790, 95% CI: 0.290–1.290, $p = 0.002$) were significant predictors of increasing IR, as measured by HOMA, in these GDM offspring but not HDL-chol ($p = 0.753$) or glucose tolerance ($p = 0.457$). WC and TGs remained significant factors after adjustment for child's age, gender, family history of diabetes, and presence of metabolic syndrome in the mother (beta: 0.845, 95% CI: 0.125–1.564, $p = 0.018$ and beta: 0.678, 95% CI: 0.130–1.225, $p = 0.005$, respectively). Fasting glucose is not included in the model as it is used to calculate HOMA.

There were no significant differences between the presence and the absence of metabolic markers of IR in offspring when related to their birthweight (difference in medians: 15.5, 95% CI: −355–309, $p = 0.850$).

13995448, 2008, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1399-5448.2007.00258.x. By Cochrane Peru, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

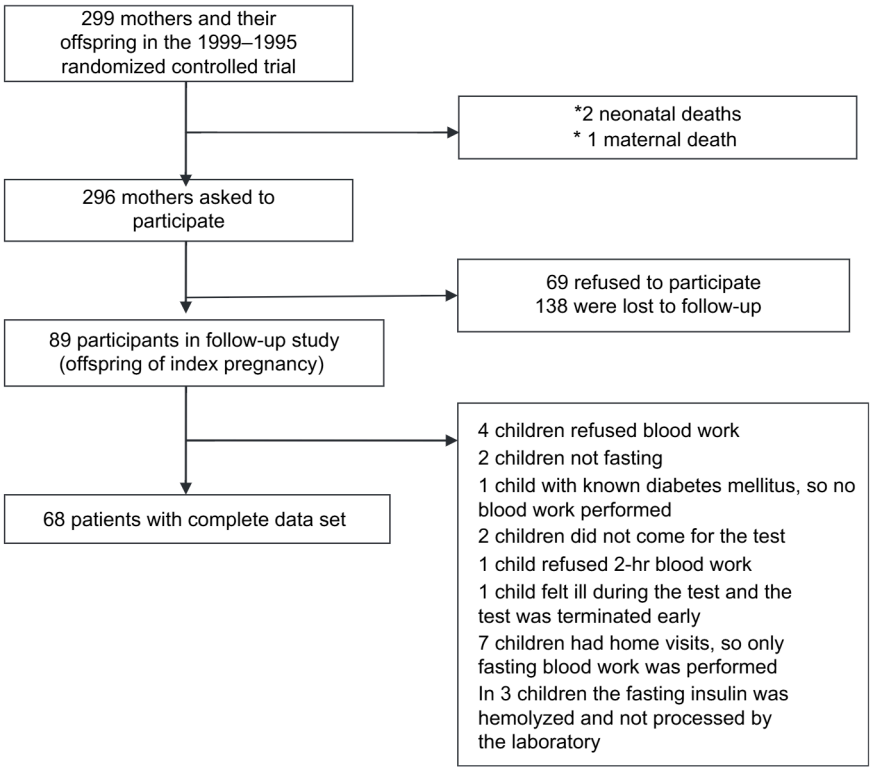


Fig. 1. Flow diagram of study participants
*None of the deaths was related to diabetes.

When LGA babies, defined as >90% for Canadian population (n = 17), were compared with AGA babies (n = 51), there was no significant difference in the presence of metabolic markers of IR (difference in proportions: −2.0, 95% CI: −28.0–20.6, p = 1.000) or HOMA levels ±2.5 (difference in proportions: −19.6, 95% CI: −44.0 to 0.0, p = 0.106).

Offspring metabolic markers of IR were compared with maternal values. There was significant correlation between offspring and maternal HDL-chol, TG, WC, and HOMA but not in fasting or 2-h plasma glucose (Table 3). There was no significant relationship between good maternal median preprandial glucose (difference in proportions: 1.6, 95% CI: −18.0 to 24.4, p = 1.000) or postprandial glucose (difference in proportions: −5.3, 95% CI: −27.6 to 18.9, p = 0.797) control and offspring with metabolic markers of IR and those without.

Discussion

Among children and adolescents, there is an increasing epidemic of obesity, metabolic syndrome, and T2DM. The goal of early detection of IR/metabolic syndrome in children and adolescents to prevent progression to cardiovascular disease, diabetes, and possibly polycystic ovary disease is an important one (2). In our low-risk cohort of Caucasian children, 34% aged 7–11 yr (mean 9.1 yr), exposed to GDM, had at least one metabolic marker of IR. Furthermore, hypertriglyceridemia and WC were found to be significant predictors of increasing IR in these GDM offspring.

It is difficult to compare studies on prevalence of metabolic markers of IR because of differences in the population and definition of metabolic syndrome selected. Obese, Hispanic children, mean age 10.9 yr, had a 90% prevalence of at least one marker of metabolic syndrome, with 30% having at least three markers (29). Caucasians aged 12–19 yr (n = 646) who participated in third National Health and Nutrition Examination Survey (NHANES III) had a prevalence of 1.5% impaired fasting glucose, 23.3% HDL <1.03 mmol/L, 23.4% TG >1.69 mmol/L, and 4.9% with hypertension (20). The rates were much higher in the overweight subgroup. Goodman et al. studied 749 Caucasian adolescents aged 12–19 yr and noted 57.7% with low HDL (<1.03 mmol/L for males and 1.29 mmol/L for females), 7.6% with elevated TG (>2.30 mmol/L), and 1.2% with impaired fasting glucose (30). There are no other studies of which we are aware that describe the metabolic indicators of IR that we have in low-risk, preadolescent offspring of mothers with GDM. The lower prevalence of markers observed in our study is likely a reflection of the young age of our subjects. With increasing age, especially at the time of puberty, IR increases and thus metabolic markers of IR would be expected to be more prevalent.

13995448, 2008, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1399-5448.2007.00258.x. By Cochrane Peru, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Characteristics, including original treatment group, of participating mothers and offspring

Characteristics	Tight control (n = 43)	Minimal intervention (n = 25)	Total n = 68
Mothers			
Follow-up age (years), mean (SD)	41 (4)	41 (4)	41 (4)
Age at delivery (years), mean (SD)	32 (4)	31 (4)	31 (4)
Ethnicity, n (%)			
Caucasian	41 (96)	21 (84)	62 (9)
Black	0 (0)	2 (8)	2 (3)
East Indian	1 (2)	1 (4)	2 (3)
Other	1 (2)	0 (0)	1 (1.5)
Did not answer question	0 (0)	1 (4)	1 (1.5)
Family history of diabetes mellitus, n (%)	24 (59)	15 (60)	39 (59)
Prepregnancy weight (kg), median (range)	63.5 (51.5–108)	70.3 (52.2–159)	65.7 (51.5–159)
Weight gain in pregnancy (kg), mean (SD)	7.3 (7.2)	3.9 (7.8)	5.8 (7.6)
Follow-up BMI (kg/m^2), median (range)	27.3 (19.4–50.5)	29.6 (21.3–49.1)	27.6 (19.4–50.5)
Waist circumference (cm), median (range)	87.6 (64.0–131.4)	92.4 (65.8–114.6)	87.8 (64.0–131.4)
Fasting glucose (mmol/L), median (range)	5.4 (4.4–7.8)	5.5 (4.8–17.6)	5.5 (4.4–17.6)
2-h glucose post-75-g CHO (mmol/L), mean (SD)	7.0 (2.6)	8.3 (2.1)	7.8 (2.4)
Triglycerides (mmol/L), median (range)	1.4 (0.7–6.5)	1.5 (0.4–5.0)	1.4 (0.4–6.5)
HDL-chol (mmol/L), median (range)	1.4 (0.78–3.25)	1.16 (0.77–1.64)	1.26 (0.77–3.25)
LDL cholesterol (mmol/L), mean (SD)	3.2 (0.8)	3.3 (0.8)	3.2 (0.8)
Total cholesterol (mmol/L), mean (SD)	5.4 (0.8)	5.3 (0.9)	5.3 (0.8)
HOMA, mean (SD)	3.36 (2.9)	4.34 (3.9)	3.7 (3.3)
Offspring			
Age (years), mean (SD)	9.0 (0.8)	9.3 (0.8)	9.1 (0.8)
M : F	20:23	10:15	30:38
Gestational age, median (range)	38.4 (28.7–41.7)	39.3 (36.0–41.1)	38.9 (28.7–41.7)
Birthweight (g), mean (SD)	3296 (682)	3679 (569)	3437 (665)
Large-for-gestational-age babies (>90% for Canadian population), n (%)	8 (19)	9 (36)	17 (25)
Waist circumference (cm), median (range)	59.5 (50.8–84.9)	60.1 (51.3–90.3)	59.6 (50.8–90.3)
BMI, n (%)			
<85th percentile	32 (74)	17 (68)	49 (72)
85th to 95th percentile	5 (12)	3 (12)	8 (12)
>95th percentile	6 (14)	5 (20)	11 (16)
Fasting glucose (mmol/L), mean (SD)	5.0 (0.4)	4.9 (0.4)	4.9 (0.4)
Fasting insulin (pmol/L), median (range)	44 (24–184)	52 (7–108)	47 (7–184)
2-h glucose post-glucose load (mmol/L), mean (SD)	5.9 (1.2)	5.9 (0.83)	5.9 (1.1)
Total cholesterol (mmol/L), mean (SD)	4.2 (0.5)	4.4 (0.8)	4.2 (0.6)
Triglycerides (mmol/L), median (range)	0.80 (0.4–2.7)	0.83 (0.5–5.4)	0.82 (0.4–5.4)
LDL cholesterol (mmol/L), mean (SD)	2.4 (0.5)	2.52 (0.9)	2.42 (0.7)
HDL-chol (mmol/L), mean (SD)	1.4 (0.3)	1.3 (0.3)	1.4 (0.3)
HOMA, mean (SD)	1.6 (0.9)	1.7 (0.8)	1.6 (0.9)

BMI, body mass index; CHO, carbohydrate; F, female; HDL-chol, high-density lipoprotein cholesterol; HOMA, homeostasis model assessment; LDL, low-density lipoprotein; M, male.

The most consistent feature across all studies investigating markers of metabolic syndrome is that glucose abnormalities are the least common among all indicators of IR (20, 29, 30). Despite this, most screening strategies are based on measuring glucose levels. In light of our results and others, consideration should be given to studying other markers of IR in both mothers and offspring to predict offspring risk.

Other studies have investigated the relationship of maternal hyperglycemia to IR in offspring. Silverman et al. compared 168 children at ages 5–9 yr and 88 children ages 10–16 yr born to women with pregestational or gestational diabetes to an age-matched but not BMI-matched control group of healthy children (31). They found an IGT rate of 5.4 and 19.3% for GDM offspring aged 5–9 and 10–16 yr, respectively, and 2.5% in the control group. Their population included higher risk ethnic groups than ours (49% Caucasian, 23% African American, 19% Hispanic, and 9% other). However, not all studies have shown increase risk to offspring born to women with GDM. Whitaker et al. compared 58 offspring of women with diet-treated GDM to 257 offspring whose mothers had negative glucose screen at ages 8–10 yr. There was no increased risk of childhood obesity in these non-Hispanic, White offspring. The increased risk of obesity and by extension hyperinsulinemia may depend on the severity and form of the maternal diabetes (32).

13995448, 2008, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1399-5448.2007.00258.x. By Cochrane Peru, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. Metabolic markers in offspring categorized by BMI

Characteristic	BMI of <85th percentile (n = 49), n (%)	BMI of 85th–95th percentile (n = 8), n (%)	BMI of >95th percentile (n = 11), n (%)	Total n = 68, n (%)
Impaired fasting glucose	0 (0)	0 (0)	0 (0)	0 (0)
Impaired glucose tolerance	2 (4)	1 (13)	0 (0)	3 (4)
Type 2 diabetes	0 (0)	0 (0)	0 (0)	0 (0)
Triglycerides > 1.24 mmol/L	4 (8)	4 (50)	6 (56)	14 (21)
High-density lipoprotein <1.03 mmol/L	2 (4)	4 (50)	1 (9)	7 (10)
Waist circumference >90th percentile	0 (0)	0 (0)	6 (55)	6 (8)
Homeostasis model assessment >2.5	2 (4)	3 (38)	5 (45)	10 (15)

BMI, body mass index.

If there is an association between maternal hyperglycemia and IR in offspring, it could be related to birthweight, development of childhood obesity, and/or exposure to hyperglycemia *in utero per se*. Some (33) but not all (31) studies have linked offspring IR to birthweight. We did not show any relationship between birthweight and offspring IR and maternal glucose control as evidenced by capillary glucose measurements or prevalence of offspring IR. The increased risk for obesity and the metabolic syndrome for children exposed to hyperglycemia *in utero* may not be a modifiable risk factor as there is no evidence yet that maternal glycemic control will reduce this lifetime risk (11). However, early identification of these individuals with earlier screening may allow for lifestyle or pharmaceutical intervention known to prevent the progression from IGT to diabetes (34, 35) or the development of cardiovascular events.

Our study is limited by several factors. We did not assess blood pressure. However, IR is less tightly associated with hypertension than with other markers (29, 36). Our sample size is small but was dictated by the number of willing participants from the original study. The lack of a control group to determine prevalence of markers of metabolic syndrome in our non-GDM offspring is also a limiting factor in determining the risk associated with GDM. We did not feel that it would be feasible to perform the necessary investigations on children not at risk, especially given that we only had a 30% enrollment rate from individuals previously committed to a clinical trial, with some vested interest in the long-term follow-up of their children.

Table 3. Correlation between maternal and offspring metabolic markers of insulin resistance

Variable	Correlation coefficient	p Value for correlation coefficient
High-density lipoprotein	0.51	<0.001
Triglyceride	0.27	0.024
Homeostasis model assessment	0.42	<0.001
Waist circumference	0.34	0.004
Fasting Sugar	0.16	0.190
2-h oral glucose tolerance test	0.03	0.780

In summary, the long-term health of children exposed to GDM *in utero* may include an increased risk of IR and metabolic syndrome. The potential for intervening in children at risk with lifestyle changes and possibly medications, before progression to cardiovascular disease or diabetes, is an important one. Our current screening strategies, focusing on FBG measurement in children, may need to be reconsidered if we are to identify these individuals early. Whether offspring outcomes are preventable with maternal glucose control is not clear. Further longitudinal studies, of sufficient sample size, should be performed to determine if offspring risk can be reduced with tight maternal glycemic control and to develop appropriate screening strategies for these children.

Acknowledgements

This work was supported from a grant from Physician Services Incorporated. The authors thank Dr Peter Garner who initiated the original study on which this work is based. His premature death prevented him from seeing the impact his work has had on the offspring of women with GDM.

References

1. Bloomgarden ZT. Type 2 diabetes in the young. The evolving epidemic. Diabetes Care 2004: 27: 998–1010.
2. Ten S, Maclaren N. Insulin resistance syndrome in children. J Clin Endocrinol Metab 2004: 89: 2526–2539.
3. National Cholesterol Education Program (NCEP) Expert Panel on Detection. Evaluation and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III). Third Report of the National Cholesterol Education Program (NCEP) Expert Panel on Detection, Evaluation, and Treatment of High Blood Cholesterol in Adults (Adult Treatment Panel III) final report. Circulation 2002: 106: 3143–3421.
4. Alberti KG, Zimmet PZ. Definition, diagnosis and classification of diabetes mellitus and its complications.

13995448, 2008, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1399-5448.2007.00258.x. By Cochrane Peru, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Part 1. Diagnosis and classification of diabetes mellitus, provisional report of a WHO consultation. Diab Med 1998: 15: 539–553.
5. Einhorn D, Reaven GM, Cobin RH et al. American College of Endocrinology position statement on the insulin resistance syndrome. Endocr Pract 2003: 9: 237–252.
6. Grundy SM, Brewer HB Jr, Cleeman JI et al. Definition of metabolic syndrome: report of the National Heart, Lung and Blood Institute/American Heart Association conference on scientific issues related to definition. Circulation 2004: 109: 433–438.
7. Katzmarkzyk PT, Perusse L, Malina RM et al. Stability of indicators of the metabolic syndrome from childhood and adolescence to young adulthood: the Quebec Family Study. J Clin Epidemiol 2001: 54: 190–195.
8. Chen W, Srinivasan SR, Li S, Xu J, Berenson GS. Metabolic syndrome variables at low levels in childhood are beneficially associated with adulthood cardiovascular risk. The Bogalusa Heart Study. Diabetes Care 2005: 28: 138–143.
9. Berenson GS. Childhood risk factors predict adult risk associated with subclinical cardiovascular disease: the Bogalusa Heart Study. Am J Cardiol 2002: 90 (Suppl.): 3L–7L.
10. Dabelea D, Pettitt DJ, Hansen RL, Imperatore G, Bennett PH, Knowler WC. Birth weight, type 2 diabetes, and insulin resistance in Pima Indian children and young adults. Diabetes Care 1999: 22: 944–950.
11. Murtaugh MA, Jacobs DR, Moran A, Steinberger J, Sinaiko AR. Relation of birth weight to fasting insulin, insulin resistance, and body size in adolescence. Diabetes Care 2003: 26: 187–192.
12. Dabelea D, Knowler WC, Pettitt DJ. Effects of diabetes in pregnancy on offspring: follow-up research in the Pima Indians. J Matern Fetal Med 2000: 9: 83–88.
13. Sobngwi E, Boudou P, Mauvais-Jarvis F et al. Effect of a diabetic environment in utero on predisposition to type 2 diabetes. Lancet 2003: 361: 1861–1865.
14. Vohr B, Mcgarvey ST, Tucker R. Effects of maternal gestational diabetes on offspring adiposity at 4-7 years of age. Diabetes Care 1999: 22: 1284–1291.
15. Lindsay RS, Hanson RL, Bennett PH, Knowler WC. Secular trends in birth weight, BM and diabetes in the offspring of diabetic mothers. Diabetes Care 2000: 23: 1249–1254.
16. Canadian Diabetes Association; Clinical Practice Guidelines Expert Committee. Type 2 diabetes in children and adolescents. Can J Diabetes 2003: 27 (Suppl. 2): S91–S93.
17. Garner P, Okun N, Keely E et al. A randomized controlled trial of strict glycemic control and tertiary level obstetric care versus routine obstetric care in the management of gestational diabetes: a pilot study. Am J Obstet Gynecol 1997: 177: 190–195.
18. Malcolm JC, Lawson ML, Gaboury I, Lough G, Keely E. Glucose tolerance of offspring of mother with gestational diabetes mellitus in a low-risk population. Diabet Med 2006: 23: 565–570.
19. Hatem M, Anthony F, Hogston P, Row DJF, Dennis KJ. Reference values for 75 g oral glucose tolerance test in pregnancy. Br Med J 1988: 296: 676–678.
20. Cook S, Weitzman M, Auinger P, Nguyen M, Dietz WH. Prevalence of a metabolic syndrome phenotype in adolescents: findings from the third National Health and Nutrition Examination Survey, 1988-94. Arch Pediatr Adolesc Med 2003: 56: 821–827.
21. Fernandez JR, Redden DT, Pietrobelli A, Allison DB. Waist circumference percentiles in nationally representative samples of African-American, European-American, and Mexican-American children and adolescents. J Pediatr 2004: 145: 439–444.
22. Sinha R, Fisch G, Tegue S et al. Prevalence of impaired glucose tolerance among children and adolescents with marked obesity. N Engl J Med 2002: 346: 802–810.
23. Gomez-Diaz R, Aguilar-Salinas CA, Moran-Villota S et al. Lack of agreement between revised criteria of impaired fasting glucose and impaired glucose tolerance in children with excess body weight. Diabetes Care 2004: 27: 2229–2232.
24. Carr DB, Utzschneider KM, Hull RL et al. Intra-abdominal fat is a major determinant of the National Cholesterol Education Program Adult Treatment Panel III Criteria for the Metabolic Syndrome. Diabetes 2004: 53: 2087–2094.
25. Freedman DS, Serdula MK, Srinivasan SR, Berenson GS. Relation of circumferences and skinfold thicknesses to lipid and insulin concentrations in children and adolescents: the Bogalusa Heart Study. Am J Clin Nutr 1999: 69: 308–317.
26. Kucsmarski RJ, Ogden CL, Grummer-Strawn LM et al. CDC Growth Charts: United States Advanced Data from Vital and Health Statistics, No.314. Hyatteville, MD: National Center for Health Statistics, 2000. www.cdc.gov/nchs/data/ad/ad314.pdf.
27. Hadjiyannakis S. The metabolic syndrome in children and adolescents. J Paediatr Child Health 2005: 10: 41–47.
28. Keskin M, Kurtoglu S, Kendirci M, Atabek E, Yazici C. Homeostasis model assessment is more reliable than the fasting glucose/insulin ratio and quantitative insuin sensitivity check index for assessing insulin resistance among obese children and adolescents. Pediatrics 2005: 115: e500–e503.
29. Cruz ML, Weigensberg MJ, Huang TK, Ball G, Shaibi GQ, Goran MI. The metabolic syndrome in overweight Hispanic youth and the role of insulin sensitivity. J Clin Endocrinol Metab 2004: 89: 108–113.
30. Goodman E, Daniels SR, Morrison JA, Buang B, Dolan LM. Contrasting prevalence of and demographic disparities in the World Health Organization and National Cholesterol Education Program Adult Treatment Panel III Definitions of Metabolic Syndrome among adolescents. J Pediatr 2004;: 145: 445–451.
31. Silverman BL, Metzger BE, Cho NH, Loeb CA. Impaired glucose tolerance in adolescent offspring of diabetic mothers. Diabetes Care 1995: 18: 611–617.
32. Whitaker RC, Pepe MS, Seidel KD, Wright JA, Knopp RH. Gestational diabetes and the risk of offspring obesity. Pediatrics 1998: 101: E9.
33. Boney CM, Verma A, Tucker R, Vohr BR. Metabolic syndrome in childhood: association with birth weight, maternal obesity, and gestational diabetes mellitus. Pediatrics 2005: 115: e290–e296.
34. Chiasson JL, Josse RG, Gomis R, Hanefeld M, Karasik A, Laakso M; STOP-NIDDM Trial Research Group. Acarbose treatment and the risk of cardiovascular disease and hypertension in patients with impaired glucose tolerance: the STOP-NIDDM trial. JAMA 2003: 290: 486–494.
35. Knowler WC, Barrett-Conner E, Fowler SE et al. Diabetes Prevention Program Research Group. Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. N Engl J Med 2002: 346: 393–403.
36. Cook S, Weitzman M, Auinger P et al. Prevalence of a metabolic syndrome phenotype in adolescents: findings from the third National Health and Nutrition Examination Survey, 1988-1994. Arch Pediatr Adolesc Med 2003: 157: 821–827.

13995448, 2008, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/j.1399-5448.2007.00258.x. By Cochrane Peru, Wiley Online Library on [16/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License